

Lecture 01

What is Cloud Computing?

- On-demand availability of computer system resources, especially computing power and storage
- automation of virtualization
- Storing and accessing data and programs over the internet

The Deployment Models of the Cloud

Private Cloud:

- Cloud services used by a single organization, not exposed to the public.
- Complete control
- Security for sensitive applications
- Meet specific business needs

Public Cloud:

- Cloud resources owned and operated by a third-party cloud service provider delivered over the Internet.
- Six Advantages of Cloud Computing

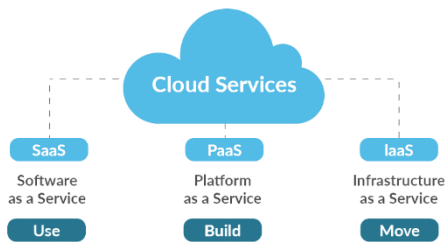
Hybrid Cloud:

- Keep some servers on premises and extend some capabilities to the Cloud
- Control over sensitive assets in your private infrastructure
- Flexibility and cost-effectiveness of the public cloud

Cloud Service Providers



Service Models of Cloud Computing



Platform as a Service (PaaS)

Tools over internet, usually for application development



Infrastructure as a Service (IaaS)

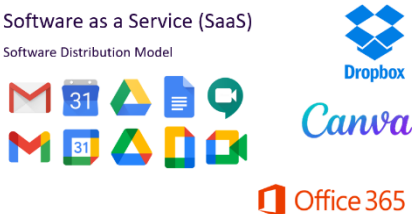
Computing Resources over internet



You've been using some Cloud services

Software as a Service (SaaS)

Software Distribution Model



Gmail

- E-mail cloud service
- Pay for ONLY your emails stored (no infrastructure, etc.)



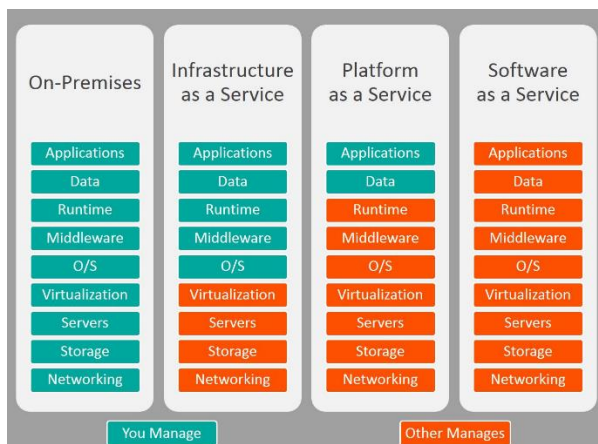
Dropbox

- Cloud Storage Service
- Originally built on AWS



Netflix

- Built on AWS
- Video on Demand



Example of Cloud Computing Types

- Infrastructure as a Service:
 - Amazon EC2 (on AWS)
 - GCP, Azure, Rackspace, Digital Ocean, Linode
- Platform as a Service:
 - Elastic Beanstalk (on AWS)
 - Heroku, Google App Engine (GCP), Windows Azure (Microsoft)
- Software as a Service:
 - Many AWS services (ex: Rekognition for Machine Learning)
 - Google Apps (Gmail), Dropbox, Zoom



Lecture 7

What is a Server?

- Servers are computers that provide “services” to “clients”
- They are typically designed for reliability and to service a large number of requests
- Organizations typically require many physical servers to provide various services (Web, Email, Database, etc.)
- Server hardware is becoming more powerful and compact

Compact Servers

- Organizations would like to conserve the amount of floor space dedicated to their computer infrastructure
- For large-scale installations, compact servers are used. This helps with:
 - Floor Space
 - Manageability
 - Scalability
 - Power and Cooling

Blades and Blade Enclosures

- A blade server is a stripped down computer with a modular design
- A blade enclosure holds multiple blade servers and provides power, interfaces and cooling for the individual blade servers

What is a Data Center?

- A data center is a facility used to house computer systems and associated components, such as networking and storage systems, cooling, uninterruptable power supply, air filters...
- A data center typically houses a large number of heterogeneous networked computer systems
- A data center can occupy one room of a building, one or more floors, or an entire building

Data Center Components

- Air conditioning
 - Keep all components in the manufacturer’s recommended temperature range
- Redundant Power
 - UPS/Generators
 - Multiple power feeds
- Fire protection
- Physical security
 - CCTV/Access Control
- Monitoring Systems
- Connectivity
 - Multiple ISPs/Leased Lines

Lecture 12

Big Data refers to extremely large volumes of data—both structured and unstructured—that cannot be easily managed, processed, or analyzed using traditional databases and tools. It is often characterized by the **5 V's**:

- **Volume**: Massive amounts of data
- **Velocity**: Speed of data generation and processing
- **Variety**: Different data types (text, images, videos, etc.)
- **Veracity**: Data accuracy and trustworthiness
- **Value**: Extracting meaningful insights from data

What Do We Do with Data?

We:

- **Store** it securely in databases or cloud storage
- **Access** it for analysis, decision-making, or application usage
- **Process** it to extract useful information, find patterns, and make predictions (e.g., through analytics or machine learning)

Requirements to Transform IT into a Service (ITaaS)

Transforming IT into a service involves turning infrastructure, software, or platforms into **on-demand, scalable, and user-friendly services**. Here are the key requirements:

Core Requirements

Requirement	Explanation
Connectivity	Reliable networking to move data efficiently across systems and locations.
Interactivity	Seamless user interfaces and APIs for smooth user-system interaction.
Reliability	High availability; failures must be minimal as they affect many users.
Performance	Services must perform well and be as fast or faster than traditional solutions.
Pay-as-you-Go	Users pay only for what they use—no upfront investment.
Ease of Programmability	Simple and flexible tools for developers to build and integrate services.
Manage Big Data	Capability to store, process, and analyze massive datasets.
Efficiency	Optimal use of resources (cost, power, space).
Scalability & Elasticity	Easily scale up/down based on demand—automatically or with minimal effort.

Example: Cloud-Based Customer Relationship Management (CRM) - Salesforce

How it meets the requirements:

- **Connectivity:** Accessible over the internet from any device.
- **Interactivity:** User-friendly dashboards, drag-and-drop UI, and robust APIs.
- **Reliability:** Uptime SLA of 99.9%, globally distributed data centers.
- **Performance:** Optimized infrastructure behind the scenes.
- **Pay-as-you-Go:** Subscription-based pricing.
- **Ease of Programmability:** Offers AppExchange and APIs for custom development.
- **Manage Big Data:** Can handle millions of customer records and interactions.
- **Efficiency:** Users don't worry about infrastructure; only pay for what they need.
- **Scalability & Elasticity:** Easily add users, features, or storage as the business grows.

Would you like a summarized infographic-style version of this table for easy use in a slide?

Lecture 14

Microsoft Azure Definition:

Microsoft Azure is a **cloud computing platform and service** offered by Microsoft that provides a wide range of **on-demand cloud services**, such as computing, storage, networking, databases, and AI, used to build, test, deploy, and manage applications through Microsoft-managed data centers.

Virtual Network (VNet) in Azure:

An **Azure Virtual Network (VNet)** is a **logically isolated network** in the Azure cloud where you can run your **virtual machines (VMs), services, and applications** securely. It is similar to a traditional on-premises network but with Azure resources.

- **Supports** subnets, routing, firewalls, VPN connections, and network security groups.
- **Enables** secure communication between:
 - Azure resources
 - On-premises infrastructure
 - The internet (if configured)

Three Examples of Azure Resources:

Resource	Purpose
Virtual Machine (VM)	Compute power to run applications, OS, and services
Azure Blob Storage	Store unstructured data like files, images, videos
Azure SQL Database	Managed relational database service

Resource Group in Azure:

A **Resource Group** in Azure is a **container** that holds related resources for an application or solution. It helps in **organizing, managing, and monitoring** resources as a single unit.

- **Benefits:**
 - Simplifies billing and access control

- Allows bulk actions like deployment or deletion
- Helps logically group by lifecycle or department

Azure subscriptions Case Study

An account can have multiple subscriptions, but it's only required to have one. In a multi-subscription account, you can use the subscriptions to configure different billing models and apply different access-management policies. You can use Azure subscriptions to define boundaries around Azure products, services, and resources. There are two types of subscription boundaries that you can use:

- **Billing boundary:** This subscription type determines how an Azure account is billed for using Azure. You can create multiple subscriptions for different types of billing requirements. Azure generates separate billing reports and invoices for each subscription so that you can organize and manage costs.
- **Access control boundary:** Azure applies access-management policies at the subscription level, and you can create separate subscriptions to reflect different organizational structures. An example is that within a business, you have different departments to which you apply distinct Azure subscription policies. This billing model allows you to manage and control access to the resources that users provision with specific subscriptions.

Types of Azure Subscription Boundaries Identified in the Case Study

The case study highlights **two types of Azure subscription boundaries**:

1. Billing Boundary

- **Purpose:** Separates **cost and billing** for different departments, projects, or environments.
- **Use in Case Study:**
 - *"You can create multiple subscriptions for different types of billing requirements."*
 - Each subscription gets **separate billing reports and invoices**, helping to **organize and manage costs**.
- **Example:** A company might create:
 - One subscription for **development**
 - Another for **production**
 - A third for **testing**, each with its own budget

2. Access Control Boundary

- **Purpose:** Enables setting different **access-management policies** per subscription.
- **Use in Case Study:**
 - *"You can create separate subscriptions to reflect different organizational structures."*
 - Useful for **isolating departments** or teams by giving them separate control over their resources.
- **Example:**
 - **HR, Finance, and IT** departments each get their own subscription with role-based access tailored to their needs

On What Basis Should You Create Azure Subscriptions?

You should create **Azure subscriptions** based on **logical separation** of environments, access control, and cost management. Here are the main bases:

1. Environment-Based Separation

Create separate subscriptions for:

- **Development**
- **Testing**
- **Staging**
- **Production**

◆ **Why?** To isolate resources, control access, and avoid accidental changes across environments.

2. Organizational Structure

Create subscriptions based on:

- **Departments** (e.g., HR, Finance, IT)
- **Teams or Business Units**

- **Projects or Clients**

◆ **Why?** Each group can have tailored permissions and resource access, ensuring better governance.

3. Billing and Cost Management

Create subscriptions to:

- Track **usage and cost** by team, product, or project
- Separate **chargebacks or budgets**

◆ **Why?** Azure bills at the **subscription level**, so this helps in monitoring and allocating expenses.

4. Compliance and Security

Create subscriptions to:

- **Isolate sensitive data**
- Meet **regulatory or compliance** needs (e.g., GDPR, HIPAA)
- Apply stricter security controls

◆ **Why?** Security and access policies can be enforced at the subscription level.

Summary Table

Basis	Purpose	Example
Environment	Isolate dev/test/prod environments	Dev subscription, Prod subscription
Organizational Unit	Reflect teams/departments structure	Finance vs. IT subscription
Billing	Track and manage costs by workload or function	Subscription per project/client
Compliance/Security	Isolate resources/data to meet security or legal needs	Separate subscription for healthcare data

